

HARSH SHAH

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SUMMARY

Data Scientist with 4+ years of experience in building data-driven solutions. Expertise in machine learning, deep learning, NLP, and computer vision, utilizing tools like SQL, Spark, MongoDB, AWS, and GCP. Improved conversion rates by 15% and ensured 90%+ data accuracy through user behavior modeling and rigorous validation frameworks. Automated ETL workflows and monitoring systems, enhancing decision-making for 30,000+ users.

EDUCATION

Boston University | Master of Science in Data Science | GPA: 3.60

Sep 2024 – Dec 2025

Fairleigh Dickinson University | Bachelor of Science in Computer Science | GPA: 3.69

Aug 2017 – May 2021

SKILLS

Data Science & Machine Learning: Deep Learning, CNNs, RNNs, Mask R-CNN, Transformer Models (GPT, BERT, BioBERT, ClinicalBERT), GANs, Sentiment Analysis, Text Classification, Summarization, Clustering, Retrieval-Augmented Generation (RAG)

Data Engineering & ETL: Data Modeling, Data Warehousing, ETL Pipelines, Data Integration, Apache Kafka, Apache NiFi

Cloud Platforms: Azure (ML Studio, Databricks, Synapse, Data Factory, Event Hubs, Blob Storage), AWS, GCP, Spark, Hadoop

Data Visualization & BI: Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis, Hypothesis Testing, A/B Testing, Time Series Forecasting, ANOVA, Regression, Bayesian Inference

Databases: SQL, Query Optimization, Database Management, Data Mining, Data Quality

Programming & Scripting: Python (Pandas, NumPy, SciPy), SQL, R, Bash/Shell, JavaScript

MLOps & DevOps: Model Monitoring, Model Versioning, CI/CD, Docker, Kubernetes, Terraform, Ansible, Jenkins

WORK EXPERIENCE

Petpooja (Restaurant POS Software)

Jun 2025 – Jul 2025

Data Science Intern

Ahmedabad, India

- Enhanced analytics dashboards (sales, inventory, taxation, staffing), improving reporting speed by 40%.
- Built aggregation logic for city/outlet-level data, boosting monitoring efficiency across 100+ outlets.
- Modeled user order flows and behavior, informing UI/UX updates that improved conversions by 15%.
- Explored ML-based tax categorization methods, reducing reporting errors by 25%.

General Motors

Jul 2022 – Jun 2024

Data Scientist

Warren, MI

- Performed time series and statistical analysis on customer and vehicle data to improve forecasting accuracy and insights for 30k users.
- Developed Excel dashboards, KPI trackers, and pivot table reports to summarize data and support faster, data driven decisions.
- Built and evaluated machine learning models for user/vehicle usage estimation using data from secure database pipelines.
- Applied advanced analytical techniques to enhance customer experience insights and support strategic business decisions.
- Implemented data-quality checks, validation rules, and governance controls to ensure accurate and secure data across systems.

General Motors

Jun 2021 – Jun 2022

Data Analyst

Warren, MI

- Validated and analyzed data outputs from 400+ API endpoints supporting enterprise analytics systems, ensuring 90%+ data accuracy in production reports.
- Created and maintained 100+ test cases and validation scenarios in JIRA to verify data pipelines, dashboards, and ML model outputs.
- Assisted in data quality checks, anomaly detection, and reconciliation to improve reliability of analytics workflows.
- Supported senior analysts by documenting validation processes and data issues, helping streamline reporting and analysis cycles.
- Mentored and collaborated with junior QA trainees on data validation and analysis best practices, improving team efficiency by 25%.

Silicon IT Hub Pvt Ltd.

Sep 2020 – Jan 2021

Android Application Developer Intern

Ahmedabad, India

- Designed an Android app to identify nearby truck parking facilities using GPS, improving logistics efficiency, and reducing downtime.

PROJECTS

Financial Stock Recommendation App | [App](#) | [Code](#)

Mar 2025 – May 2025

- Built a Streamlit app combining sentiment analysis (Twitter, news), fundamentals, and technical indicators.
- Designed a scoring engine producing Buy/Hold/Sell signals based on user-weighted models.
- Created dashboards to compare stocks across multiple dimensions, enabling deeper financial insights.

Predicting Respiratory Hospitalizations from Air Pollution | [Code](#) | [Dashboard](#)

Mar 2025 – May 2025

- Built machine learning models to predict hospitalization risks based on real-time air quality data.
- Designed ETL pipelines using Azure Data Factory & Event Hubs to ingest historical & real-time pollution data.
- Performed time-lag correlation analysis using Distributed Lag Models (DLMs) & Generalized Additive Models (GAMs).
- Created a Power BI dashboard for visualizing high-risk zones & sending alerts via Azure Logic Apps.

Cricket Analytics ML Integrated Platform | [Code](#)

Feb 2025 – Apr 2025

- Conducted EDA on player-level cricket data (Pandas, Seaborn) to identify trends and outliers.
- Built clustering models (KMeans, cosine similarity) to segment players by performance.

Exploratory Data Analysis of Boston's District 7's Economic and Population Development | [Code](#) | [Dashboard](#)

Oct 2024 – Dec 2024

- Empowered the Policy Director with interactive visualizations to assess decade-long socioeconomic and displacement trends.
- Built a live Python dashboard (GeoPandas, Seaborn, Tableau) comparing District 7 metrics to city averages for key indicators.